# Ashish Rana


M.Sc. Data Science, 4+ Years Data Science and Research Experience

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scholar.google.com/citations?user=5XxfLlIAAAAJ


## Education

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|---|---|
| GPA: 2.4 (Good) | *M.Sc., Data Science, University of Mannheim, Germany (2021-2023), Gut in German Scale, ≅ to USA 3.1* |
| CGPA: 9.51/10.00 | *BE, Comp. Eng., Thapar Institute of Engineering and Technology, India (2015-2019), ≅ to DE 1.3 and USA 3.8* |
| Percentage: 91% | *$12^{th}$, Army Public School, Meerut Cantt., Uttar Pradesh, India (2013-2014)* |
| GPA: 9.8/10.0 | *$10^{th}$, Army Public School, Jalandhar Cantt., Punjab, India (2011-2012)* |

## Professional Experience

**March 2026 – June 2025**
**Research Engineer, NEC Laboratories Europe (NLE), Heidelberg, Full-Time**
- Researching long-term memory in agentic LLMs and causal prototype-mechanism explanations.
- Developing scalable real-time LLM systems for emergency call handling with FastAPI and CI/CD.

Long Term Memory | Causal Prototypes | System Design and Software Releases | Supervising Thesis and Leadership

**May 2025 – January 2024**
**Research Associate, NEC Laboratories Europe (NLE), Heidelberg, Full-Time**
- Built SotA interpretable deep variants of Classification-by-Components approach.
- Developed interpretable health assistant web application with GLVQ and GTLVQ approaches.

Prototype-based Learning | Prototype Building | Software Development

**September 2023 – April 2023**
**Working Student, Institute for Enterprise Systems, Mannheim, Part-Time**
- Built data preprocessing and pipeline code optimizations for the BERT system pipeline.
- Developed novel architecture improvements for the smart meter time-series data.

Time-series Data | Transformers | Data Processing

**December 2023 – April 2023**
**Working Student, trinamiX GmbH, Ludwigshafen, Part-Time**
- Developed data processing and code optimizations for the CNN based deep learning pipeline.
- Benchmarked transfer learning performance analysis of EfficientNet with dynamic input size biometric data.

PyTorch | Hydra | MLflow

**February 2023 – September 2022**
**Working Student, FARO Technologies, Korntal-Münchingen, Part-Time**
- Conducted large scale high fidelity point cloud segmentation *($10^7$ points)* with RandLA-Net.
- Finetuning RandLA-Net on synthetically generated data and benchmarking the improvements.

PyTorch | Open3D-ML | Synthetic Data Generation

**August 2022 – June 2022**
**Data Science Intern, ZEW, Mannheim, Full-Time**
- Developed web scrapers to get multi-lingual data and built translator wrappers for syncing datasets.
- Conducted football competency statistical analysis with Difference-in-Difference models.

Selenium | Beautiful Soup | HuggingFace | PyTorch

**September 2021 – August 2019**
**Data Engineering Specialist, GE Healthcare, Bengaluru, Full-Time**
- Developed predictive outlier detection POC for MRI modality from multiple sensor data sources.
- Built machine utilization metrics to measure extra healthcare system load for COVID-19 patients.
- Developed and deployed terabyte scale ingestion *PL/pgSQL* procedures on AWS Redshift.

Python | R | PL/pgSQL | AWS | Redshift

**June 2019 – January 2019**
**Data Engineering Intern, American Express, Bengaluru, Full-Time**
- Developed Prometheus and Grafana based monitoring and alerting tools for Couchbase DB VMs.
- Designed, developed and deployed parallel VM patching scripts to reduce downtime by 4 times.

Couchbase | Python | Grafana | Prometheus

**July 2018 – June 2018**
**Software Engineering Intern, GE Digital, Bengaluru, Full-Time**
- Added SAML re-authentication module and migrated to REST architecture for Enovia PLM codebase.
- Conducted case study of open-source static analysis tools on Enovia Java codebase.

ReST | Java | SAML

## Selected Research Work

Rana, A., Hung, C.C., Sun, Q., Kunkel, J.M. and Lawrence, C., 2026. Oblivion: Self-Adaptive Agentic Memory Control through Decay-Driven Activation. arXiv preprint arXiv:2604.00131. *Pre-print Available at arXiv, ACL ARR 2026 May Submission*

arxiv.org/abs/2604.00131 nec-research.github.io/oblivion/

Built a cognitively-inspired memory control framework that casts forgetting as decay-driven reductions in accessibility, not explicit deletion.

Agentic Memory | Long-term Memory | Memory Management

Rana, A., Shaker, A., Saralajew, S., Suzuki, T., Yasuda, K., Kato, S., Wada, T., Fujikawa, T. and Kikutsuji, T. (2025) 'Prototype-Based Learning for Healthcare: A Demonstration of Interpretable AI', in Proceedings of the Demo Track of the IEEE International Conference on Data Mining Workshops 2025 (ICDMW 2025). *Best Demo Paper Award, ICDM Demo Track 2025*

10.1109/ICDMW69685.2025.00323 arxiv.org/abs/2601.02106

Built SotA system for interpretable disease risk prediction and explanation, and further used prototype-specific autoencoders to model inter-feature dependence for intervention corrections.

Prototype-based Learning | Interventions and Recommendations

Saralajew, S., Rana, A., Villmann, T. and Shaker, A. (2025) 'A Robust Prototype-Based Network with Interpretable RBF Classifier Foundations', in Proceedings of the AAAI Conference on Artificial Intelligence, Vol. 39, No. 19, pp. 20273-20282. *AAAI 2025*

github.com/nec-research/enlaight github.com/si-cim/cbc-aaai-2025 arxiv.org/abs/2412.15499

Built deep Classification-by-components architectures, conducted in depth interpretability analysis and developed software toolkits for reproducibility.

Classification-by-components | Prototype-based Learning | SoTA Interpretability Benchmark Performance

Rana, A., Oesterle, M. and Brinkmann, J. (2024) 'GOV-REK: Governed Reward Engineering Kernels for Designing Robust Multi-Agent Reinforcement Learning Systems', in Proceedings of the 23rd International Conference on Autonomous Agents and Multiagent Systems (AAMAS '24). Richland, SC: International Foundation for Autonomous Agents and Multiagent Systems, pp. 2429-2431. *Extended Abstract, AAMAS 2024*

github.com/arana-initiatives/gov-rek-marls arxiv.org/abs/2404.01131

For faster learning used reward distribution priors and Hyperband algorithm to efficiently engineer agent rewards in an automated manner for sparse Multi-Agent Systems (MAS).

Centralized and Decentralized MAS Training | Gaussian Kernel Distributions | Training and Hyperparameter Optimization

Rana, A., Golchha, P., Juntunen, R., Coajă, A., Elzamarany, A., Hung, C.C. and Ponzetto, S.P. (2022) 'LeviRANK: Limited query expansion with voting integration for document retrieval and ranking: notebook for the touché lab on argument retrieval at CLEF 2022', in CEUR Workshop Proceedings, Vol. 3180, pp. 3074-3089. RWTH Aachen. *Touché IR Workshop, CLEF 2022*

github.com/softgitron/LeviRank https://ceur-ws.org/Vol-3180/paper-259.pdf

Built novel query relevance proportionality based retrieval and implemented "Expando-Mono-Duo" design pattern with T5 models.

BM25 and Semantic retrievals | Relevance Feedbacks | Pointwise & Pairwise Ranking

Rana, A., Khanna, D., Ghosal, T., Singh, M., Singh, H. and Rana, P.S. (2022) 'RerrFact: Reduced Evidence Retrieval Representations for Scientific Claim Verification', in Proceedings of the Scientific Document Understanding Workshop at AAAI 2022. CEUR Workshop Proceedings, Vol. 3164. *SDU Workshop, AAAI 2022*

github.com/RerrFact arxiv.org/abs/2202.02646

Re-designed multi-stage information retrieval & classification modules with loosely coupled model training across different stages for maximum precision performance throughput.

Scientific Claim Verification | Modular Pipeline Design | Transformer Language Models

Rana, A., Singh, H., Mavuduru, R., Pattanaik, S. and Rana, P.S., 2022. Quantifying prognosis severity of COVID-19 patients from deep learning based analysis of CT chest images. Multimedia Tools and Applications, 81(13), pp.18129-18153. *MTAP Journal 2022*

doi.org/10.1007/s11042-022-12214-6

Designed severity estimator Single Shot Detector (SSD) pipeline for better COVID-19 prognosis on CT slice image dataset. (DOI: 10.1007/S11042-022-12214-6 )

Single Shot Detector (SSD) | Image Similarity Networks | Multiclass classification

Rana, A. and Malhi, A. (2021) 'Building safer autonomous agents by leveraging risky driving behavior knowledge', in 2021 International Conference on Communications, Computing, Cybersecurity, and Informatics (CCCI). IEEE, pp. 1-6. *Informatics Best Paper Award, CCCI 2021*

https://ieeexplore.ieee.org/document/9583209

Improved and estimated the causal agent performance improvements by utilizing collision scenario information with a causal experiment design *(DOI: 10.1109/CCCI52664.2021.9583209)*.

Model-free Deep Reinforcement Learning | Stochastic Control | Causal Experimentation Design

## Selected Project Achievements

**liveProject: Reinforcement Learning for Self-driving Vehicles** AVAILABLE @LIVEPROJECT.MANNING.COM

github.com/ashishrana160796/prototyping-self-driving-agents
Developed agent prototyping hands-on practice project for Manning Publications and made contributions to *highway-env* open-source package.

Stochastic Control | Policy Gradient Methods | Actor-Critic Methods

**Automating Couchbase Monitoring Integrations with Prometheus & Grafana** *Couchbase Connect.ONLINE 2020*

github.com/ashishrana160796/cautious-integration-analysis youtu.be/T3wVPRBY8H8?si=g20hHKjgJMM5u8Bn
Presented an orchestrating multi-tool setup with Prometheus, Couchbase-Exporter & Grafana for Couchbase monitoring with REST HTTP servers. Also, making the architectures highly available & raising predictive alerts to reduce application downtime.

ReST | Couchbase | Forecasting & Outlier Detection

## References

**Dr. Ammar Shaker**
*Senior Research Scientist*, NLE
@ Ammar.Shaker@neclab.eu

**Dr. Dr. Sascha Saralajew**
*Senior Research Scientist*, NLE
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## Languages

English (C1) ●●●●●
German (B1) ●●●○○

## Interests

- Reliable Agentic LLMs
- AI Interpretability
- Multimodal Representation Learning